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PROJECTS HISTORY

Project: Traditional RAG – AI-Powered Personal Portfolio Assistant

Organization: Personal Project (Integrated within Portfolio Website)

Technologies Used: PHP | MySQL | ChromaDB | Vector Store | OpenAI API | Agentic AI Framework | HTML5 | JavaScript | CSS3 | Python (Integration Layer) | VS Code

Challenges:

- Needed an intelligent, interactive way to showcase professional experience, achievements, and technical expertise directly on a personal portfolio site.
- Static portfolio content offered limited user engagement and could not respond contextually to visitor queries.
- Required a system capable of providing personalized, conversational answers about professional details while maintaining data accuracy and context.

Solution Provided:

- Designed and deployed a **Retrieval-Augmented Generation (RAG)**-based AI assistant integrated into the personal portfolio website.
- Implemented **vector-based knowledge retrieval** using **ChromaDB / Vector Store**, enabling context-aware responses from professional data sources (resume, project documents, achievements).
- Integrated **Agentic AI** framework for dynamic conversational handling and multi-turn dialogue management.
- Built a **floating chatbot UI** with persistent context, offering real-time interactions for visitors exploring the portfolio.
- Used **OpenAI APIs** for natural language understanding and response generation, supported by **PHP** and **MySQL** backend integration.
- Optimized performance and response accuracy through fine-tuned retrieval pipelines and memory management.

Outcome and Benefits:

- Delivered an interactive, AI-driven experience that provides **instant, contextually accurate answers** about professional background and expertise.
- Enhanced portfolio engagement, allowing visitors to explore skills, experience, and achievements conversationally.
- Demonstrated hands-on experience in **LLM orchestration, RAG architecture, and vector database integration**.
- Strengthened personal brand visibility by showcasing end-to-end AI implementation skills within a live environment.
- Established a reusable framework for future expansion into intelligent, self-learning personal assistants.

Project: AI-Driven SQL Automation and Unit Testing Framework (MCP Server Initiative)

Organization: Launch Consulting Group – Client: HealthEdge Software

Technologies Used: SQL Server 2019 | T-SQL | Python | OpenAI API | GitHub Copilot Agent | Visual Studio Code | TeamCity | Bitbucket | SourceTree | Docker (Windows Containers) | SQL Profiler | GitBash
Dec 2024 – Present.

Challenges:

- Manual SQL procedure development and testing consumed significant engineering time and increased the risk of inconsistencies.
- Unit test creation for stored procedures and functions lacked automation, leading to longer QA cycles.
- Difficulty in tracing procedure dependencies and validating business logic through standard testing practices.
- Limited integration between development, testing, and CI/CD environments reduced efficiency and visibility.

Solution Provided:

- Designed and implemented a **Model Control Protocol (MCP) Server**, enabling **natural language to SQL procedure generation** using AI Agents.
- Developed a **custom T-SQL unit testing framework** capable of automatically validating stored procedures, functions, and views.
- Integrated **GitHub Copilot Agent** in Visual Studio Code for AI-assisted test case creation and automated SQL pattern recognition.
- Used **SQL Profiler Trace** to map dependencies and functional flows for accurate test coverage.
- Established **TeamCity-based CI integration** to run unit tests across multiple environments, ensuring continuous validation.
- Built **containerized Oracle and SQL environments using Docker** to support dynamic connectivity and Test Data Management (TDM).

Outcome and Benefits:

- Reduced manual SQL coding and unit testing effort by **over 60%** through AI automation.
- Improved defect detection and code reliability by **40% during pre-UAT cycles**.
- Enabled continuous integration and faster delivery pipelines with **TeamCity automation**.
- Decreased QA setup time by **50%** using containerized, self-validating test environments.
- Significantly improved developer productivity and overall testing consistency across projects.

Project: Actuarial Data Visualization and Reporting System

Organization: TechTriad – Client: Prudential Financial (NJ)

Technologies Used: Oracle BI | SQL Developer | SQL | PL/SQL | Oracle BI Publisher | Data Modeling | Performance Tuning

Sep 2021 – Oct 2024

Challenges:

- Legacy actuarial reporting systems were inefficient, lacked visualization, and required manual effort for data preparation and validation.
- Limited alignment between actuarial teams and IT led to inconsistent report requirements and slower delivery cycles.
- Complex actuarial calculations and multi-source data models caused performance bottlenecks in existing reporting processes.

Solution Provided:

- Partnered with actuarial and business units to understand reporting requirements and streamline data visualization needs.
- Designed and developed **interactive Oracle BI dashboards** to deliver accurate and real-time actuarial insights.
- Utilized **SQL Developer** for direct SQL query optimization and integration with Oracle BI reports.
- Documented end-to-end reporting workflows, ensuring clear specifications for design, development, and testing.
- Conducted **User Acceptance Testing (UAT)** with business users to validate data accuracy, visualization, and usability.
- Optimized reporting workflows and query performance, improving data retrieval efficiency by **25%**.

Outcome and Benefits:

- Delivered a **centralized actuarial reporting system** with improved visualization and self-service analytics capabilities.
- Reduced manual reporting effort and turnaround time, enhancing operational efficiency.
- Improved decision-making through accurate, real-time insights and streamlined data workflows.
- Strengthened collaboration between actuarial and technical teams through structured documentation and testing processes.

Project: Reporting System Modernization and BI Migration

Organization: The Evolvers Group – Clients: Energy Transfer & City of Arlington (TX)

Technologies Used: SQL Server 2016 | SSRS | T-SQL | SQL Server Management Studio (SSMS) | Crystal Reports | Data Modeling | Query Optimization

June 2016 – June 2021

Challenges:

- Legacy **Crystal Reports** systems lacked scalability, automation, and flexibility for modern reporting needs.
- Inefficient SQL queries and absence of standardized report structures resulted in longer generation times and inconsistent outputs.
- Business users required enhanced data visibility through interactive, drill-down reports and better performance analytics.

Solution Provided:

- Led the **migration of 100+ Crystal Reports to SQL Server Reporting Services (SSRS)** with improved design, formatting, and parameterization.
- Developed **SSRS reports** (tabular, matrix, and drill-through) enabling faster data access and deeper insight.
- Optimized SQL Server queries and implemented indexing strategies to enhance database performance.
- Collaborated with stakeholders to gather requirements, ensuring reporting solutions aligned with business goals.
- Provided training and documentation to end-users and developers for smoother adoption and maintenance.
- Established SQL development best practices for reusability, performance, and maintainability.

Outcome and Benefits:

- Reduced report generation time by **40%** through query optimization and SSRS enhancements.
- Improved database performance and reliability with indexing and tuning strategies.
- Delivered modern, interactive reports improving stakeholder visibility into business operations.
- Enhanced data-driven decision-making through timely, accurate, and visually rich reports.
- Standardized the reporting framework, reducing future maintenance and development time.

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Sr. Application and Reporting Expert

C3i HealthCare Connections

India, Hyderabad

May 2013 – March 2016

Project Title: Crystal Reports XIR2 Maintenance and .Net Projects with SharePoint Migration

Technologies Used: Crystal Reports, MS SQL, .Net Framework, SharePoint

Project Description:

In a multifaceted role at C3i, I managed and maintained Crystal Reports, developed and documented end-user report requirements, and migrated legacy reports to Interaction Level Queries (ILQ) using MS SQL Stored Procedures. Additionally, I worked on various .Net projects, including client web portals and internal systems, and participated in the migration and administration of SharePoint portals.

Responsibilities:

Crystal Reports XIR2 Maintenance and Automation

- Management of report requests assigned Via internal “IT Request System” along with additional responsibility of web portals.
- Point of contact for C3i’s Operations Teams for the development and documentation of end user report requirements (both new and existing)
- Responsible for the maintenance of several Crystal Reports and documentation library of the code/ logic used to develop Crystal reports
- Involved in migrating Old Crystal reports to ILQ(Interaction level queries) and providing Business logic to Reporting team for preparing SSRS reports. ILQs are maintained in the terms of MS SQL Stored Procedures.
- Participate in identifying new ways to automate reporting processes for time saving, accuracy, and cost reductions.

.Net Framework Client and Internal Projects

- Worked on several client and internal projects using .net framework such as Merck Grants, ITRS (Ticketing Requesting System) web portal, Employee management System (web portal)
- Worked on several Client web portals (such as, XBiotech, Merck, Novonordisk etc.) as per the requirement.

SharePoint portal migration and Administration

- Involved in migrating SharePoint 2010 (existing portal) to SharePoint 2013, Configuration and Administration.

Sr. Client Reporting Data Expert
C3i HealthCare Connections
India, Hyderabad
May 2012 – May 2013

Project Title: Crystal Reports Development and MS SQL Optimization

Technologies Used: IT Request System, Crystal Reports, MS SQL Server, T-SQL, Stored Procedures

Project Description:

The project involved managing and developing Crystal Reports, maintaining documentation, and optimizing SQL Stored Procedures. The key focus was on ensuring timely delivery of report requests through the IT Request System, improving the performance of SQL queries, and mentoring the back-office report development team.

Responsibilities:

Report Request Management and Documentation:

- Managed report requests assigned via the internal "IT Request System," ensuring timely delivery of Crystal Reports.
- Maintained a comprehensive documentation library of the code used to develop and enhance Crystal Reports, ensuring ease of reference and reuse.

SQL Procedure Optimization and Performance Tuning:

- Provided optimization and performance tuning for MS SQL Stored Procedures to ensure reports were generated efficiently and accurately.
- Enhanced SQL queries and T-SQL code to improve overall database performance and reporting speed.

Team Training and Mentorship:

- Provided ongoing support, training, and mentorship to the back-office report development team on Crystal Reports development and MS SQL Stored Procedures, including best practices in query optimization and performance tuning.

Requirements Gathering and Application Enhancement:

- Collaborated with Ops and Client teams to gather requirements and develop enhancements for existing reporting applications, ensuring the system met evolving business needs.

Sr. Manager - Development and Analytics
C3i HealthCare Connections
India, Hyderabad
June 2011 – May 2012

Project Title: Crystal Reports Development for Clinical, Horizon Call Center, Service Ware, and Aporopos

Technologies Used: Crystal Reports, PL/SQL, Oracle, SQL Server

Project Description:

This project focused on the development of new Crystal Reports for Clinical, Horizon Call Center, Service Ware, and Aporopos, using PL/SQL procedures and Oracle as the backend. The aim was to improve report efficiency, maintain a structured documentation library, and implement automated solutions to reduce manual effort.

Responsibilities:

Crystal Reports Development:

- Developed new Crystal Reports for key applications, including Clinical, Horizon Call Center, Service Ware, and Aporopos, leveraging PL/SQL procedures with Oracle as the backend.
- Ensured reports were accurate, timely, and met the needs of business stakeholders.

Documentation Maintenance:

- Created and maintained a comprehensive documentation library of the code used in the development of Crystal Reports, ensuring smooth knowledge transfer and ease of future maintenance.

Requirements Gathering and Application Enhancements:

- Partnered with Operations and Client teams to gather detailed requirements for report enhancements and new reporting functionalities, ensuring the application was continuously improved based on user feedback and business needs.

Process Automation and Optimization:

- Participated in initiatives to identify and implement new automation solutions to streamline reporting processes, enhance accuracy, and achieve significant time and cost savings.

Manager - Development and Analytics

C3i HealthCare Connections

India, Hyderabad

June 2008 – June 2011

Project Title: Report Management, Data Warehouse Integration, and Process Automation

Technologies Used: Lotus Notes, Crystal Reports, MS SQL Server, MS Access, T-SQL, PL/SQL, Data Warehouse, Snowflake

Project Description:

This project focused on managing report requests through the Lotus Notes ticket application, integrating with the Data Warehouse, and automating various reporting templates. The objective was to streamline reporting processes, enhance accuracy, and reduce operational costs.

Additionally, the project involved maintaining and improving custom tools and applications such as the eHOC application and QC Tool, while ensuring data consistency through Snowflake integration.

Responsibilities:

Report Request Management and Data Warehouse Integration:

- Managed report requests generated via Lotus Notes ticketing system, ensuring alignment with data **warehouse** solutions such as **Snowflake** for seamless reporting.
- Acted as the main point of contact for C3i's Operations Teams, handling the development and documentation of end-user report requirements for both new and existing reports, integrated with data.

Report Development, Automation, and Data Extraction:

- Developed and troubleshoot macros, code, and automated report templates from data stored in MS SQL Server, and MS Access as requested by clients or internal operations.
- Automated and optimized existing report templates, sourcing data from the Data Warehouse, leading to improved accuracy and time savings.

Tool Maintenance and Data Consistency:

- Maintained and enhanced custom tools like the eHOC application and back-end databases, ensuring data consistency across platforms, including Snowflake and SQL Server databases.
- Gathered requirements from Ops and Client teams to enhance applications such as the Grant Training Tracker, leveraging data from the Data Warehouse for real-time reporting.
- Team Support and Training:
- Provided ongoing support, training, and mentorship to the Back Office report development team on Crystal Reports, T-SQL, PL-SQL, and Data Warehouse concepts such as Snowflake integration.

Process Automation and Improvements:

- Identified new opportunities to automate reporting processes by leveraging Data Warehouse solutions for time savings, increased data accuracy, and reduced operational costs.

Achievements:

- Successfully automated several reporting templates, using data from the Snowflake Data Warehouse, significantly reducing manual effort and improving data accuracy.
- Enhanced custom tools integrated with Snowflake, leading to more efficient processes for both internal and client teams.
- Provided training and mentorship that improved the technical skills of the Back Office team and ensured a smooth integration with Snowflake.

Sr. Developer Analytics
C3i HealthCare Connections
India, Hyderabad
January 2006 – June 2008

Project Title: Application and Report Template Development, Troubleshooting, and Post-Production Support
Technologies Used: MS SQL Server, MS Access, Excel Macros, Crystal Reports

Project Description:

This project focused on the development, implementation, and support of custom applications, report templates, and tools. The aim was to enhance in-house and third-party tools, address scalability, concurrency, and availability challenges, and provide end-user training and technical documentation.

Responsibilities:

Application and Report Template Development:

- Developed, implemented, and enhanced various applications, report templates, Excel Macros, and MS Access programming solutions to streamline business processes.
- Automated report templates and troubleshooting logic for optimizing report generation.

Documentation and Technical Writing:

- Created detailed process and technical documentation for all in-house and third-party tools, including application code and logic to ensure consistency and future maintenance ease.
- Established and maintained a documentation library for macros, coding, and system enhancements.

Solutions for Scalability, Concurrency, and Availability:

- Provided solutions for critical issues related to scalability, concurrency, and availability across various systems, ensuring system reliability and performance.

Custom Tools Maintenance and Support:

- Managed and enhanced custom tools, including eHOC application, MS SQL Server Stored Procedures, MS Access back-end database, QC Tool, and others, providing ongoing post-production support.
- Administered MS SQL Server, MS Access, and Crystal Reports for maintaining operational efficiency and resolving technical issues.

End-User Support and Training:

- Delivered training sessions and technical documentation to users on the functionality and usage of tools, including AMS, Tech Scorecard, Grading Tool, and Bonus Payout Tool.
- Acted as a key point of contact for all application-related queries and enhancements from the Operations and Client teams.

Crystal Report Specialist
C3i HealthCare Connections
India, Hyderabad
January 2005 – January 2006

Project Title: Dynamic Crystal Reports Prototyping and Ad-Hoc Query Development

Technologies Used: Crystal Reports, MS Access, SQL

Project Description:

Developed dynamic Crystal Reports prototypes and ad-hoc queries based on client business requirements, while maintaining and troubleshooting existing reports.

Responsibilities:

- Built dynamic Crystal Reports prototypes from client requirements and sample design templates.
- Developed ad-hoc queries for both client needs and internal reporting support, tracking report generation statistics.
- Created a front-end Crystal Lite Web-page dashboard, enabling users to select and run reports from multiple databases like cmsholding, horizonP, and ServicewareP.

PHP Programmer

Net Solutions India

India, Nasik

March 2004 – January 2005

Project Title: Website Development and Maintenance

Technologies Used: PHP, MySQL, CMS, Payment Gateway Integration

Project Description:

Developed and maintained multiple websites, including newtrendmosaic.com, flywire.com.au, webdesigners.net.au, and phoenixtraining.info, focusing on CMS integration and payment gateway systems.

Responsibilities:

- Gathered requirements from clients to develop specific modules for websites.
- Maintained and updated existing web pages based on requests from the Project Manager.
- Implemented and enhanced CMS in the admin sections for easier content management.
- Integrated payment gateway systems into client websites to support online transactions.